

The Science of Can and Can't

a from-zero crash course in constructor theory

You do not need any mathematics or any physics to read this. You need only the patience to meet a few new ideas one at a time. By the end you will be able to say, in your own words, what constructor theory is; how it differs from the ordinary way physics has worked for three hundred years; what its handful of core ideas mean; what it has so far been used to talk about — information, life, and heat; and — equally important — how much of it is settled and how much is still an open bet by a small group of physicists. That last part matters, and I will not hide it: this is a young idea, not a finished one.

We are going to do the whole thing in one kitchen. Recipes, cooks, ingredients, and the dishes you can and can't make: that is the only picture you will need to hold, and we will build everything else on top of it.

Chapter 1 — *A puzzle you already know*

Crack an egg into a hot pan and stir. In a minute you have scrambled eggs. Now try to run it backwards: take the scrambled eggs and turn them back into a whole, unbroken egg with the yolk sitting neat inside the white. You can't. Nobody can. Not a clumsy cook, not a brilliant one, not a cook with a billion-dollar laboratory. The forward recipe is easy; the backward recipe is, as far as anyone can tell, flatly impossible.

Here is a second one. Write a recipe on a card — *two eggs, a pinch of salt, thirty seconds in the pan* — and you can copy that card perfectly, by hand, as many times as you like. Every copy works as well as the original. But the cake itself, the finished dish, never copies itself and never un-bakes. The *recipe* copies freely; the *cake* does not.

So nature is fussy. It hands out some transformations for free and refuses others outright, and the pattern of what it allows and what it forbids is not random — it is lawful, dependable, the same in every kitchen on Earth. Step back and a strange question appears, the question this whole book is about:

What if the deepest laws of physics were not predictions of what happens next, but statements about which transformations are possible and which are impossible — and why?

That is the wager behind constructor theory. It was proposed by the physicist David Deutsch in a 2013 paper, and it has been developed since then by Deutsch together with Chiara Marletto at the University of Oxford.¹ It is not a crank's daydream and not settled textbook science either; it is a serious research programme, made by serious people, that is still trying to prove its worth. Marletto wrote a popular book about it with a title that captures the whole idea in four words: *The Science of Can and Can't*.²

Hold on to the egg and the recipe card. We will keep coming back to them, because the egg you can't unscramble and the recipe you can copy turn out to be two of the deepest facts in physics — and constructor theory is the attempt to put facts of exactly that shape at the very bottom of the stack.

Recap: nature permits some transformations and forbids others, lawfully. Constructor theory asks whether those permissions and prohibitions — the can's and can't's — should be the foundation of physics itself.

¹ David Deutsch's foundational paper "Constructor Theory" appeared in the journal *Synthese* in 2013; he and Chiara Marletto have developed the programme at the University of Oxford since around 2012 (F1, F2).

² Marletto's *The Science of Can and Can't* was published in 2021 (F3).

Chapter 2 — How physics normally works

To feel what is new about constructor theory, you have to first see clearly the older picture it wants to join — the one physics has run on since Newton. Call it the **prevailing conception**, because that is what it is: the standard, reigning way of doing fundamental physics.

It works like this. Take any physical system — a planet, a billiard ball, a galaxy, a single atom. Write down two things. First, its **initial conditions**: where everything is right now and how fast it is moving. Second, its **laws of motion**: the rules that say how those positions and speeds change from one instant to the next. Feed both into the mathematics, turn the handle, and out comes a prediction of what the system will do at every future moment.³

It is worth pausing on how staggeringly well this works. It is how we land probes on comets, predict eclipses years ahead to the second, and design every bridge and engine ever built. The prevailing conception is one of the most successful ideas in human history. Constructor theory is not trying to throw it out. Keep that firmly in mind: nobody is claiming Newton was wrong.

Think of the prevailing conception as a **cosmic prediction machine**. You give it the present — the exact state of the kitchen,

³ This is the “prevailing conception” of fundamental physics: predict what will happen from a system’s initial conditions plus its laws of motion (F6).

every ingredient in its place — and it tells you the future, dish by dish, instant by instant. Brilliant. But notice what kind of question it is built to answer. It answers *“given this starting point, what happens next?”* It is a machine for grinding the present forward into the future.

And here is the crack that constructor theory pushes into. Some of the most interesting things in the world are awkward to phrase as “given this start, what happens next?” Two of them in particular kept nagging at Deutsch: the physics of **information** and the physics of **life**.⁴ Think about a fact like *“this message can be copied”* or *“this recipe can be passed on accurately.”* That is not really a statement about one particular system following one particular trajectory from one particular starting point. It is a statement about what is *possible* — about a whole sweep of would-be transformations, most of which never actually happen. The prediction machine, for all its power, has no natural slot for a claim like that. It tells you what *does* happen. It is oddly quiet about what merely *could*.

That quiet is the opening. The next chapter walks through it.

Recap: the prevailing conception predicts the future from initial conditions plus laws of motion, and does so superbly — but it is built to

⁴ A stated motivation for constructor theory is that the physics of information and of life are awkward to capture in the language of trajectories and initial conditions (F8).

*say what happens, not what is possible, and
possibility is exactly where information and life
live.*

Chapter 3 — The science of can and can't

Here is the move. Constructor theory proposes that we state the laws of physics **only** in terms of which transformations are possible and which are impossible, and why.

⁵ Not “given this start, here is the future,” but “this kind of change can be made happen; that kind never can.”

A statement about what *could* or *could not* happen — rather than what actually does — has a name: a **counterfactual**. The word means *contrary to fact*: a claim about a state of affairs that may never come to pass. “I could have caught the earlier train” is a counterfactual; you didn’t, but you could have. Constructor theory’s central and most radical proposal is that counterfactuals like these — *can* and *can’t* — belong at the very foundation of physics, as basic facts about the world, not as afterthoughts.⁶ That is the whole reason for the book title: it really is meant to be a *science of can and can’t*.

If putting “can’t” at the bottom of physics sounds exotic, here is the reassuring part: physics has secretly done it for centuries. Take one of the most famous laws there is, the conservation of energy — energy is never created or destroyed. There is an exactly equivalent way to say the same thing, and it is a flat statement of impossibility: **you cannot build a perpetual-**

⁵ Constructor theory expresses physical laws exclusively in terms of which transformations (“tasks”) are possible versus impossible, and why (F7).

⁶ Constructor theory deliberately admits counterfactual statements — about what could or could not happen — into fundamental physics (F14, F15).

motion machine. You cannot build the free-lunch kitchen that cooks forever out of nothing.⁷ That “cannot” is not a soft engineering caution; it is as hard as any law in science. So the idea of a deep law that takes the form “*this is impossible*” is not strange at all. Constructor theory’s bet is that this shape — the impossible, the forbidden recipe — is not the exception but the rule, the form *every* fundamental law should take.

There is a lovely twist in this way of thinking, which Marletto puts crisply: “Declaring something impossible leads to more things being possible.”⁸ It sounds backwards, so here is the kitchen version. The moment you accept, as an ironclad law, that no kitchen can cook from nothing, you suddenly know an enormous amount about every real kitchen: every dish must be paid for in ingredients and heat, every process must balance its books. A firm “can’t” does not shrink the world; it carves it into a definite shape, and a world with a definite shape is one you can actually reason about. Rule out the impossible cleanly, and the possible comes into focus.

The two ways of doing physics now sit side by side. It is worth seeing them in one view:

⁷ Conservation of energy is equivalent to the statement that a perpetual-motion machine is impossible — physics already hides can’t-statements at its foundations (F16).

⁸ Marletto’s phrasing of a guiding heuristic of the programme (F9).

	The prevailing conception	Constructor theory
What's fundamental	initial conditions + laws of motion	which tasks are possible vs impossible
What a law says	"given this start, here is what happens next"	"this transformation can be made to happen; that one never can — and here's why"
Its native question	what <i>does</i> happen?	what <i>can</i> and <i>can't</i> happen?
A familiar example	predict a planet's orbit from its position and speed	a perpetual-motion machine is impossible

The same physics, two foundations: the standard picture makes the next moment fundamental; constructor theory makes the possible/impossible divide fundamental (F6, F7, F13, F16).

The right-hand column is the whole programme in one box. Everything else is working out how to speak that language precisely. For that, we need exactly three words.

Recap: constructor theory states laws as counterfactuals — what can and can't happen — and that is less strange than it sounds, because conservation of energy already is the impossibility of perpetual motion. Ruling out the impossible sharpens what's possible.

Chapter 4 — The three words: *task*, *constructor*, *substrate*

To say “this transformation is possible” precisely, you need a word for the transformation, a word for the thing that does it, and a word for the thing it is done to. Three words. We already have perfect kitchen names for all three.

A task is a recipe. In constructor theory, a **task** is the abstract specification of a transformation: *these* input attributes turned into *those* output attributes.⁹ It is exactly what a recipe is — *raw eggs and salt go in; scrambled eggs come out*. A task does not say who does it or how; it only says what counts as having been done. It is the *what*, not the *how*. Notice it is a specification, a description on a card, not an event: the recipe for a soufflé exists whether or not anyone is cooking one tonight.

A constructor is a cook. A **constructor** is the thing that actually carries the task out — and here is the one feature that makes the word do real work: a constructor performs the task *and remains able to perform it again*.¹⁰ A cook scrambles your eggs and is still, a minute later, perfectly able to scramble the next person’s. An oven bakes one loaf and stands ready for the next. The cook is not used up by the cooking. The technical comparison physicists reach for is a **catalyst**

⁹ A task is the abstract specification of a transformation as input–output pairs of attributes (F10).

¹⁰ A constructor is an entity that can cause a task to occur while retaining the ability to cause it again (F11).

— in chemistry, something that makes a reaction go while itself emerging unchanged, ready to do it again.¹¹ A constructor is anything with that catalyst-like quality: it drives a change and survives intact to drive it once more.

A substrate is the ingredients. The **substrate** is the physical stuff the constructor works on — the system whose attributes get transformed.¹² The eggs are the substrate; the cook is the constructor; “scramble them” is the task.

With those three words, the central distinction of the whole theory becomes easy to state. A task is **possible** if the laws of physics allow a constructor for it to exist — even if nobody has built that constructor yet. A task is **impossible** if a law of physics forbids it from ever being done with arbitrarily high accuracy, no matter how clever the would-be cook.¹³ “Scramble an egg” is a possible task: cooks for it exist by the million. “Unscramble an egg” is, as far as physics tells us, an impossible task: no cook for it can exist, ever. That is the difference between the two recipes we started with in Chapter 1, now stated in the theory’s own terms.

Notice how much weight the word “possible” is carrying, because this is where the new language earns its keep. In ordinary speech “possible” often means “someone has

¹¹ The catalyst comparison is the one Marletto uses: something that drives a change while remaining able to do it again (F11).

¹² A substrate is the physical system a constructor acts on, whose attributes are transformed (F12).

¹³ A task is impossible if a law of physics forbids its being done with arbitrarily high accuracy; otherwise it is possible — a constructor for it could in principle exist (F13).

done it.” Here it means something stronger and stranger: a task is possible if the laws of physics *permit a constructor for it*, whether or not anyone has ever built one, or ever will. There are surely dishes no cook has yet invented — recipes that have never once been made anywhere in the universe — which are nonetheless perfectly possible, only waiting for a clever enough cook. The prediction machine of Chapter 2 has nothing to say about a dish that has never been cooked; there is no trajectory to compute, no event to predict. But constructor theory does have something to say: it classes that uncooked dish firmly on the *possible* side of the line, right alongside the egg you can scramble, and far away from the egg you cannot unscramble. A law that can speak about transformations that have never happened — and may never happen — is doing something the older picture cannot.

That is genuinely all the machinery there is. Before we put it to work, let me show you the single piece of notation in this entire book — and as promised, I will name every symbol in it.

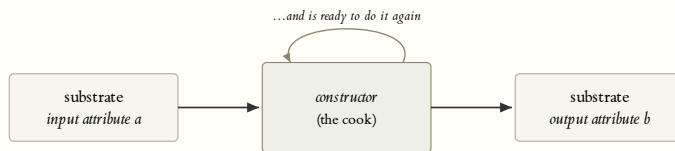
The math, gently

Physicists like to write a task compactly.
They might write a task as:

$$a \rightarrow b$$

Read it left to right. The letter *a* stands for the input — the attribute the substrate starts with (our raw eggs). The arrow $\rightarrow$ means "can be transformed into." The letter *b* stands for the output — the attribute the substrate ends with (scrambled eggs). So the whole line reads: "the task of turning *a* into *b*." That's it. No calculation, no hidden steps — it is shorthand for a recipe, with *a* the ingredients and *b* the dish. Whenever you meet it, say "*a* becomes *b*" and you have read it correctly.

It helps to see the three words working together as one picture. A constructor sits in the middle; the substrate goes in one side carrying its input attribute and comes out the other carrying its output attribute; and crucially the constructor is still standing afterwards, ready for the next round:



Recap: a task is a recipe (input $\rightarrow$ output), a constructor is a cook who does it and can do it again — like a catalyst — and a substrate is the ingredients. A task is possible if a constructor for it could exist, impossible if a law forbids it.

A constructor transforms a substrate from input attribute a to output attribute b and remains able to do it again (F10, F11, F12).

Chapter 5 — Information as physics

Now the payoff begins. The reason Deutsch and Marletto wanted this language is that it lets them talk cleanly about things the prediction machine handled awkwardly — and the first of those is **information**.

What is information, physically? We have a strong everyday sense of it: a message, a recipe card, the text of this book. The deep feature of information is that it can be **copied**. You can take a recipe card and produce a second card that says exactly the same thing, and then a third, each as good as the first. Now hear that sentence in the theory's own words: *copying a recipe is a task, and it is a possible task*. “Take a card bearing message X and produce a second card also bearing X” — that is a transformation with an input and an output, and physics permits a constructor for it. Information, in this view, *is* the kind of attribute that certain copying-and-transforming tasks are possible for.

In 2015 Deutsch and Marletto published exactly this idea, in a paper called “Constructor theory of information,” in a leading physics journal.¹⁴ Its boldest claim is a reversal of the usual order of things. We normally think of information as a pure, abstract idea — something from mathematics or logic that exists before and

¹⁴ Deutsch & Marletto, “Constructor theory of information,” *Proceedings of the Royal Society A* (2015) (F17).

above the physical world, and that physical things merely happen to carry. The constructor theory of information turns that around: it says information is **not** an abstract given at all. What information is, and how it behaves, is *fixed by the laws of physics* — specifically, by which copying and transforming tasks those laws make possible or impossible.¹⁵ Information is physical to the bone.

That sounds abstract, so make it concrete in the kitchen again. Ask: *which copying tasks does nature actually permit?* “Take a card reading **2 eggs, a pinch of salt** and produce a second card reading the same” — permitted, easily; you do it with a pencil. The whole edifice of writing, printing, and computer memory is built on the fact that this particular task is possible. But you can equally well ask about copying tasks that nature might *forbid*, and the surprising lesson of modern physics is that not everything you can imagine copying can actually be copied. The content of an information law lies exactly in that pattern — in the sharp line between the copies you can make and the ones you can’t. The constructor theory of information is the attempt to write that line down as fundamental law, rather than to assume, as we usually do, that information is some frictionless ideal that physics merely stores.

¹⁵ The paper holds that information is not an a priori mathematical or logical concept; its nature is fixed by the laws of physics — by which transformations are possible or impossible (F18).

Why go to this trouble? Because it offers a single, deep foundation for things that otherwise sit in separate boxes. The everyday information in a recipe card and the strange information inside a quantum computer look like utterly different beasts. The hope of the constructor theory of information is that both are species of one thing — both are patterns of which tasks are possible — and so both can be understood with one set of laws.¹⁶ Whether that hope is fully made good is another matter, and we will be honest about it in Chapter 8. But you can already see the shape of the ambition: take the slippery notion of information, and pin it to the floor of physics using nothing but the language of can and can't.

Recap: information is the kind of attribute that can be copied — and copying is a possible task. The 2015 constructor theory of information argues information is not an abstract idea floating above physics but something defined by the physical laws of what can and can't be copied, with one root for classical and quantum information alike.

¹⁶ The programme aims for constructor-theoretic laws underlying both classical and quantum information (F19).

Chapter 6 — *Life without a designer*

The second thing that nagged at Deutsch was **life** — and life is where the kitchen metaphor reaches its most beautiful extension.

Living things look *designed*. An eye, a wing, a cell: each is so well-fitted to its job that for most of history the only available explanation was a designer who fitted it. Darwin's great achievement was to show how the appearance of design can arise with no designer at all, through replication and natural selection: things that copy themselves, with occasional small errors, and whose better-copying versions come to outnumber the rest.¹⁷ That is not in dispute here. The question Marletto asked is a *physicist's* question, one layer underneath Darwin: **what must the laws of physics be like for that whole process to be possible in the first place?**

In ordinary kitchen terms: life is not merely a recipe, and not merely a cook. Life is a **recipe that builds the cook**. A living thing is like a recipe card which, when followed, produces a brand-new cook *and* a fresh copy of the card — a cook who can then go and do the same again. Run that loop and you have reproduction. The astonishing claim of evolution is that this self-building loop needs no designer to set it up.

¹⁷ Neo-Darwinian evolution explains the appearance of design without a designer, via replication and natural selection (F21).

Marletto's 2015 paper, "Constructor theory of life," asks whether such a loop is compatible with what she calls **no-design laws** — laws of physics that contain no special clauses for living things, no secret recipe for an eye hidden in the fundamental equations.¹⁸ Her result is clean and, I think, genuinely striking. Accurate self-reproduction, replication, and natural selection *are* compatible with no-design laws — and the only non-trivial thing the laws must allow is this: that **digital information** can be physically instantiated.¹⁹

"Digital" is the load-bearing word, so let me unpack it in the kitchen. A digital recipe is one written in discrete, exact symbols — 2 eggs, not *a generous glug*; 200 °C, not *fairly hot* — so that it can be copied perfectly, with no drift creeping in across thousands of copies. A vague, "analogue" recipe degrades a little each time it is handed on by word of mouth, until the dish is unrecognisable. A digital one does not.

You can feel the difference with a line of cooks passing a recipe down the bench. Pass a soft, fuzzy instruction — *cook it till it looks about right* — from one cook to the next, and twenty cooks later it is hopeless: each one rounds it off a little, and the small errors pile up. But pass on a single sharp instruction — *exactly two eggs* — and twenty cooks later it is still *exactly two eggs*: a choice between distinct, countable symbols leaves

¹⁸ Marletto, "Constructor theory of life," *Journal of the Royal Society Interface* (2015) (F20).

¹⁹ Its central result: accurate self-reproduction, replication and natural selection are compatible with no-design laws, the only non-trivial requirement being that the laws allow digital information to be physically instantiated (F22).

no room for the errors to accumulate. That is the entire magic of “digital,” and it is why a recipe written in exact symbols can be copied ten thousand times without drifting, while a recipe shouted across a noisy kitchen cannot survive ten retellings. Life needs that drift-free exactness, and life has it: this is what the genetic code, written in the discrete letters of DNA, actually provides. The deep point is that the *possibility of exact, digital copying* is the one special gift the laws of physics must offer for open-ended life to get going. Grant that one thing, and no designer is required for any of the rest.

The paper adds one more piece worth knowing, because it sharpens the picture. A mere **replicator** — the bare recipe card — is not enough on its own. To be copied accurately, a replicator needs a **vehicle**: the surrounding machinery that reads it and carries the copying out. The recipe card plus the cook-and- kitchen that reads it and rebuilds it: together those two form a **self-reproducer**.²⁰ In a living cell, the DNA is the replicator and the rest of the cell’s machinery is the vehicle. Neither copies itself alone; together they close the loop.

²⁰ Under no-design laws an accurate replicator requires a vehicle that, together with the replicator, constitutes a self-reproducer (F23).

Recap: living things look designed but aren’t. Constructor theory of life shows that self-reproduction and natural selection are compatible with laws that have no special provisions for life, provided one thing — that the laws allow exact, digital information to be copied. A replicator

needs a vehicle; together they make a self-reproducer.

Chapter 7 — *The one-way kitchen: thermodynamics*

Return to the egg you can't unscramble, because it is time to give that stubbornness its proper name. The study of heat, energy, and which way processes naturally run is called **thermodynamics**, and its most famous result is the **second law of thermodynamics**: left alone, things run *down* and not back up. Heat flows from hot to cold, never the reverse; milk stirs into coffee but never out of it; the egg scrambles but never unscrambles. The kitchen, almost entirely, is a **one-way kitchen**.

You can already hear that this is a statement about can and can't. The second law, spelled out, says: *the task of running this transformation forward is possible; the task of running it backward is impossible*. It is a law about which recipes nature will and will not let you cook. That is why thermodynamics fits the constructor-theory mould so naturally — it was practically *born* speaking the language of impossibility.²¹ Recall the deepest example from Chapter 3: energy conservation is precisely the impossibility of a perpetual-motion machine (F16). Thermodynamics is full of “can't”s like that, and constructor theory's job is to state them exactly.

That precision is the actual contribution. The traditional statements of the second law

²¹ Thermodynamics fits the can/can't mould: some transformations run one way but not the reverse — the second law — stated as which tasks are possible vs impossible (F25).

come with fine print — they tend to assume large systems, or near-equilibrium conditions, or averages over many particles, and they get slippery when you ask about a single small machine. Marletto’s “Constructor Theory of Thermodynamics” aims to give the second law an *exact*, scale-independent formulation in the language of possible and impossible tasks — one that holds for a single tiny constructor as cleanly as for a steam engine.²² Whether it fully succeeds is, again, part of the open question of Chapter 8; the point for now is that the *kind* of law thermodynamics is — a law of one-way recipes — is exactly the kind constructor theory was built to express.

²² Marletto’s “Constructor Theory of Thermodynamics” aims to state the second law exactly and scale-independently in constructor-theoretic terms (F24).

The same treatment has been extended even to **probability** — the mathematics of chance and uncertainty — in a paper called “Constructor Theory of Probability.”²³ The details are beyond a first crash course, but the direction is the one you would now predict: take a notion usually imported from pure mathematics, and try to ground it in physical statements about which tasks are possible.

²³ Marletto’s “Constructor Theory of Probability” extends the approach to probabilistic statements (F26).

Recap: the second law of thermodynamics says forward transformations are possible and their reverses are impossible — a law of one-way recipes, naturally a can’t-statement. Constructor theory aims to state it exactly, at any scale, and extends the same approach to probability.

Chapter 8 — Is it true? What's settled, and what's open

I have saved the most important chapter for last, and I have built it at the same density as the first, because the easiest way to mislead you about a young idea is to let the excitement run on past the evidence. So here is the honest accounting.

First, what is genuinely real and not in doubt. The *papers exist* and have passed peer review in respectable journals: the foundational 2013 paper, the constructor theory of information (2015), the constructor theory of life (2015), and the work on thermodynamics and probability. These are real proposals by real physicists, worked out in mathematical detail. We can lay out plainly what has been built so far:

Subsidiary theory	Everyday handle	The core claim
Information (2015)	a recipe you can copy exactly	information is defined by which copying tasks physics permits, not by abstract logic
Life (2015)	a recipe that builds the cook	self-reproduction needs no designer, only laws that allow digital information
Thermodynamics	the one-way kitchen	the second law stated exactly as which tasks are possible vs impossible
Probability	—	chance grounded in physical possibility rather than pure mathematics

The same can/can't language has been applied, in published work, to information, life, and thermodynamics (F17, F19, F20, F22, F24, F25, F26).

Now, the part that intellectual honesty demands. Constructor theory is a **young research programme, not established physics**. It is not in the standard textbooks. Most working physicists do not use it, and many have not formed a view on it at all. It sits outside the current mainstream, and it has not been confirmed by experiment.²⁴ Even sympathetic coverage of the programme says the quiet part out loud: whether constructor theory “can really deliver ... remains to be seen,” and broad experimental backing is, so far, not there.²⁵ When you explain this idea to someone,

²⁴ Constructor theory is an early-stage research programme, not established or experimentally confirmed, and outside the mainstream (F27).

that sentence belongs in your explanation. It is not a defeat of the idea; it is the accurate status of the idea.

So is there *any* point where this style of thinking touches an experiment you could actually run? The most concrete one is striking, and it is worth ending on — with a careful fence around it. Marletto, working with the physicist Vlatko Vedral, proposed a real test concerning **gravity**.²⁶ The reasoning runs like this. Take two small masses and place each in a **superposition** — a genuine quantum situation in which a particle has no single definite position but is, in a precise sense, spread across more than one place at once. Let the two masses interact through nothing but their mutual gravity. If, through gravity alone, they become **entangled** — woven into a shared quantum state so that you cannot fully describe one without the other — then, Marletto and Vedral argue, gravity itself must be quantum. The crux is a clean impossibility statement of exactly the kind this book has been about: a purely *classical* go-between cannot create entanglement between two systems.²⁷ So if gravity manages it, gravity is not classical.

This test is regarded as far closer to feasible than the older dreams of quantum gravity, such as detecting an individual graviton — though “closer” still means fiercely difficult, and it has not yet been carried out.²⁸ And

²⁵ Even sympathetic coverage notes that whether constructor theory can “really deliver ... remains to be seen,” with broad empirical validation still undeveloped (F28).

²⁶ Marletto & Vedral (2017, *Physical Review Letters*) proposed that gravitationally induced entanglement between two masses would be evidence that gravity is quantum (F29).

²⁷ The crux is that a purely classical mediator cannot create entanglement between two systems — so if gravity does, gravity is not classical (F29).

here is the fence: this experiment is *inspired* by the same counterfactual, information-first style of reasoning that Marletto pursues, but it is **not** a direct test of constructor theory as a whole. If it succeeds, it tells us something profound about gravity; it does not, by itself, certify the larger programme. I draw the line there deliberately, because blurring it is exactly how a young idea gets oversold.

Step back and you can see why someone like Deutsch finds the whole project worth a career. Long before constructor theory, in his 1997 book *The Fabric of Reality*, Deutsch argued that four great strands of explanation — quantum physics, evolution, computation, and the theory of knowledge — are far more deeply connected than they look, and that you cannot fully understand any one of them without the others.²⁹

Constructor theory is, in a sense, the tool he and Marletto built to try to make good on that vision: a single language — possible and impossible tasks — general enough to talk about information, life, heat, and ordinary matter all at once. That is the dream. Whether the dream becomes physics is precisely what is still being decided.

Recap and close: the papers are real and peer-reviewed, but constructor theory is young, outside the mainstream, and not experimentally confirmed — say so when you explain it. Its most concrete contact with experiment, the

²⁸ This test is considered closer to feasible than detecting gravitons or Planck-scale effects, but remains very hard and has not yet been performed (F30).

²⁹ In *The Fabric of Reality* (1997) Deutsch argued that quantum physics, evolution, computation, and the theory of knowledge are deeply interconnected (F4, F5).

gravity-entanglement test, is a related idea, not a verdict on the whole. The ambition is a single language of can and can't deep enough to unify information, life, and matter — and that ambition is still an open bet. You can now explain all of that in your own words, which is exactly what we set out to do.

Glossary

- **Task** — the specification of a transformation: a given input attribute turned into a given output attribute. A recipe. (Notation: $a \rightarrow b$, “turn a into b .”)
- **Constructor** — something that carries out a task and remains able to carry it out again, like a catalyst. A cook, or an oven.
- **Substrate** — the physical system a constructor acts on; the thing being transformed. The ingredients.
- **Possible task** — a task that the laws of physics allow a constructor for, even if none has been built yet.
- **Impossible task** — a task that some law of physics forbids from ever being done with arbitrarily high accuracy.
- **Counterfactual** — a statement about what could or could not happen, as opposed to what actually does happen.
- **Prevailing conception** — the standard way of doing physics: predict the future from initial conditions plus laws of motion.
- **No-design laws** — laws of physics with no special provisions for living things; no design built into the fundamental rules.
- **Digital information** — information written in discrete, exact symbols, so it

can be copied perfectly without drifting;
the kind life relies on.

- **Replicator** — the thing that gets copied, on its own (e.g. a recipe card, or DNA).
- **Vehicle** — the surrounding machinery that reads a replicator and carries out the copying.
- **Self-reproducer** — a replicator together with its vehicle, forming a system that copies itself.
- **Second law of thermodynamics** — the law that, left alone, processes run one way and not the reverse (heat flows hot to cold; eggs scramble but don't unscramble).
- **Superposition** — a genuine quantum situation in which a system has no single definite value (e.g. position) but is spread across more than one at once.
- **Entanglement** — a shared quantum state of two systems so linked that neither can be fully described on its own.

Further reading

- Chiara Marletto, *The Science of Can and Can't* (2021) — the popular book by one of the programme's two principals; the friendliest full-length way in. [S4]
- “How to Rewrite the Laws of Physics in the Language of Impossibility,” *Quanta Magazine* (2021) — an excellent, careful magazine explainer with Marletto, including the honest caveats. [S6]
- David Deutsch, “Constructor Theory,” *Synthese* (2013) — the foundational paper; technical, but this is where it all starts. [S1]
- Deutsch & Marletto, “Constructor theory of information,” *Proc. R. Soc. A* (2015) — the information paper. [S7]
- Chiara Marletto, “Constructor theory of life,” *J. R. Soc. Interface* (2015) — the life paper. [S11]
- constructorththeory.org — the programme's own site, with copies of the papers and newer work. [S17]
- David Deutsch, *The Fabric of Reality* (1997) — the earlier book whose vision of four interwoven “strands” sets the stage for the whole project. [S10]